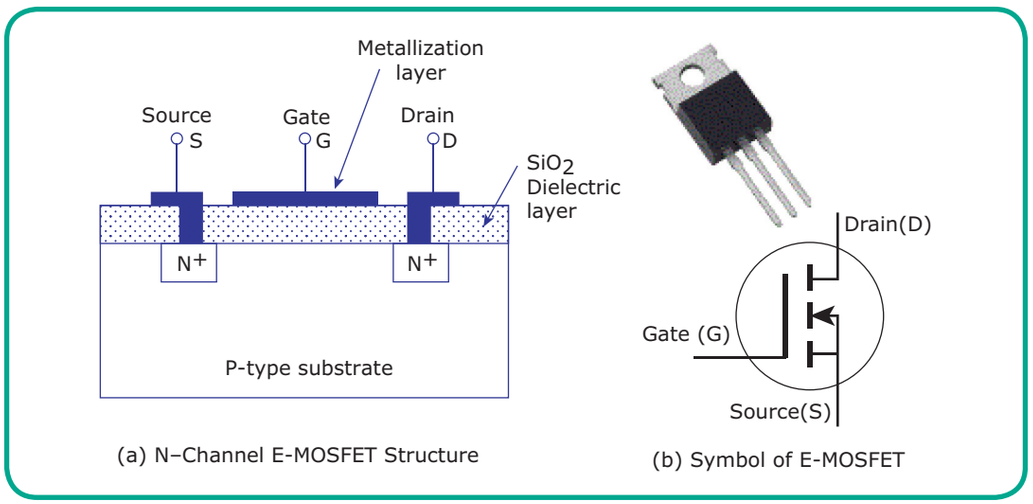



**Figure 6.11** Construction of N-Channel E-MOSFET and Its Symbol

diffused. These two *N* regions act as Source (*S*) and Drain (*D*). A thin layer of $SiO_2$ is grown over its surface then a metal contact is provided at the top of the $SiO_2$ layer, which acts as a Gate (*G*). The oxide layer provides high input impedance ($10^{10}$ Ω to $10^{15}$ Ω) to the MOSFET.

**Working Principle:** Figure 6.12 shows the normal biasing of n-channel E-MOSFET. It is always operated with positive gate to source voltage ($V_{GS}$). The drain is kept at positive potential ($V_{DS}$) with respect to the source. When $V_{GS}$ is equal to zero, the *PN* junction between the drain and the substrate is reverse biased. Hence, only very small leakage current ($I_D$) flows through the device.

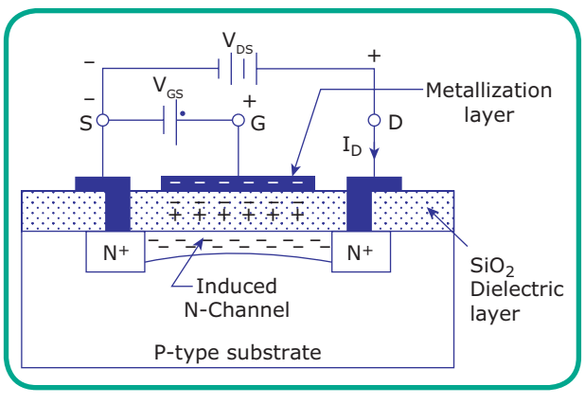



**Figure 6.12** Operation of N-Channel E-MOSFET

When gate is given a positive voltage (due to capacitor action), induced negative charges will be setup in the P-type substrate just adjacent to the $SiO_2$ layer. These negative charges, which are minority carriers in the P-type substrate form an inversion layer (N-type layer) between source and drain. This layer formed only if $V_{GS}$ exceeds a certain value called threshold voltage ($V_{GS(th)}$). When the gate to source voltage is greater than $V_{GS(th)}$, drain current ($I_D$) will flow from drain to source through the induced N-channel (Inversion layer). Thus, the drain current is 'enhanced' by the positive gate voltage and hence this device is called enhancement type MOSFET.

### 6.10.2 Depletion MOSFET (DE-MOSFET)

The depletion MOSFET can be operated in either depletion mode or enhancement mode. Therefore, it is called as DE-MOSFET. There are two types of depletion MOSFET. They are

1. N channel depletion MOSFET
2. P-channel depletion MOSFET



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